Big Data Integration & Processing

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SoICT - HUST

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Outline

- Data integration as a new abstraction
- Examples of data integration applications
- Heterogeneity
- Goal of data integration
- Data integration architectures
- Big Data Integration - Challenges

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Data Integration

- Databases are great ?
  - Assuming you've put it all into your schema.
- Data sets are often created independently
  - Only to discover later that they need to combine their data!
  - Data in different systems, different schemata and limited interfaces to data.

➢Data integration: tie together different sources, controlled by many people, under a common schema.

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DBMS: it's all about abstraction

- *Logical* vs. *Physical*; *What* vs. *How.*

Students:

SSN	Name	Category
123-45-6789	Charles	undergrad
234-56-7890	Dan	grad
	...	...

Takes:

SSN	CID
123-45-6789	CSE444
123-45-6789	CSE444
234-56-7890	CSE142
	...

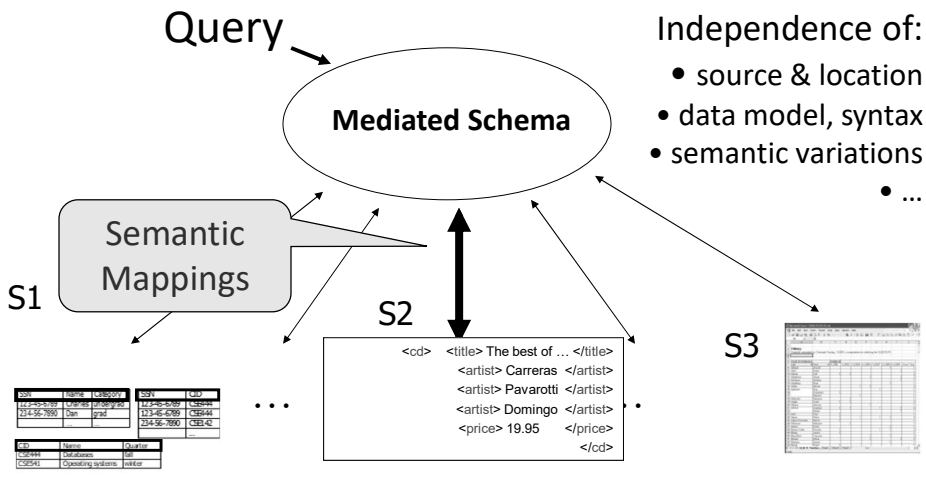
Courses:

CID	Name	Quarter
CSE444	Databases	fall
CSE541	Operating systems	winter

```
SELECT  C.name
FROM Students S, Takes T, Courses C
WHERE S.name="Mary" and
        S.ssn = T.ssn and T.cid = C.cid
```

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Data Integration: A Higher-level Abstraction



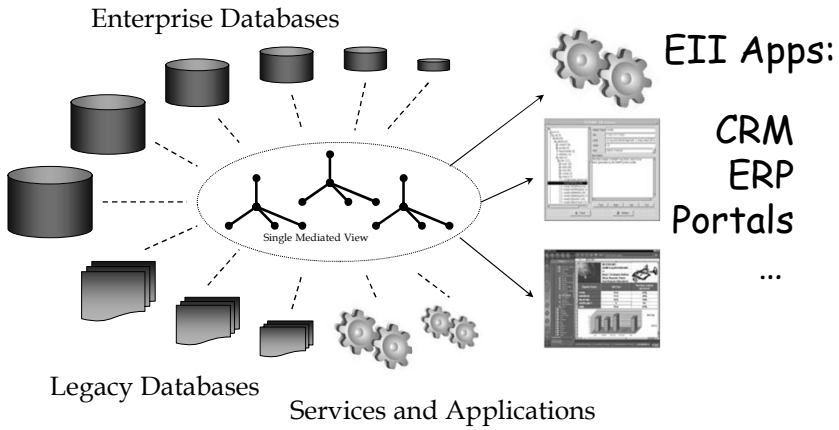
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Applications of Data Integration

- Business
- Science
- Government
- The Web
- Pretty much everywhere

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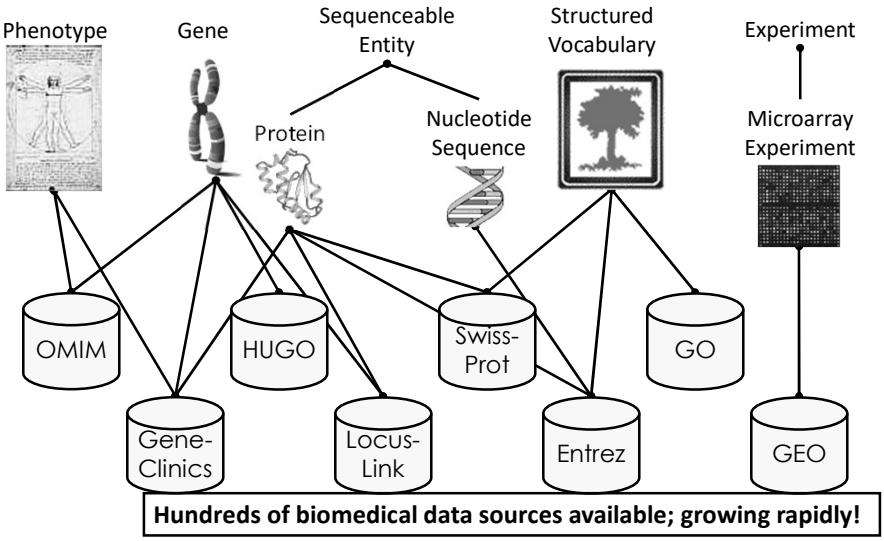
Application Area 1: Business



50% of all IT $$$ spent here!

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Application Area 2: Science



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Application Area 3: The Web



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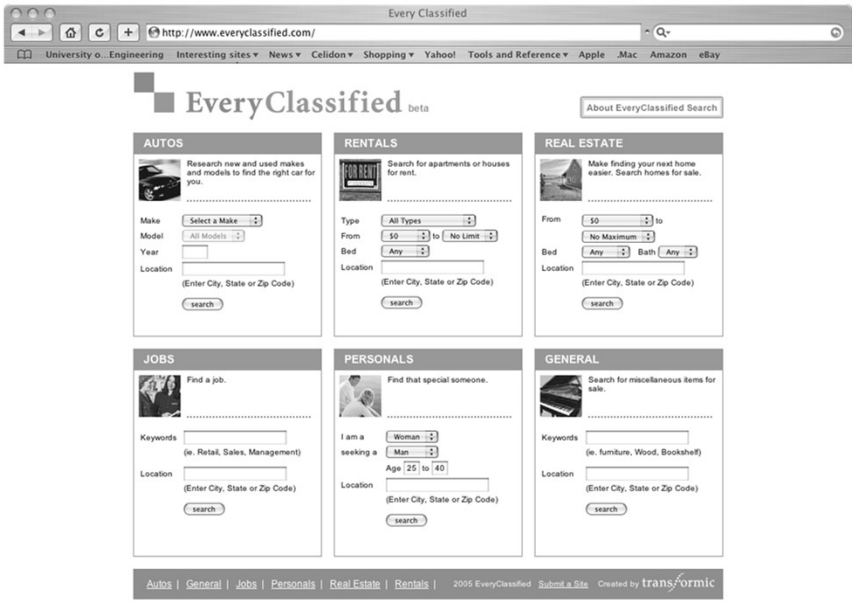
The Deep Web

- Millions of high quality HTML forms out there
- Each form has its own special interface
  - Hard to explore data across sites.
- Goal (for some domains):
  - A single interface into a multitude of deep-web sources.

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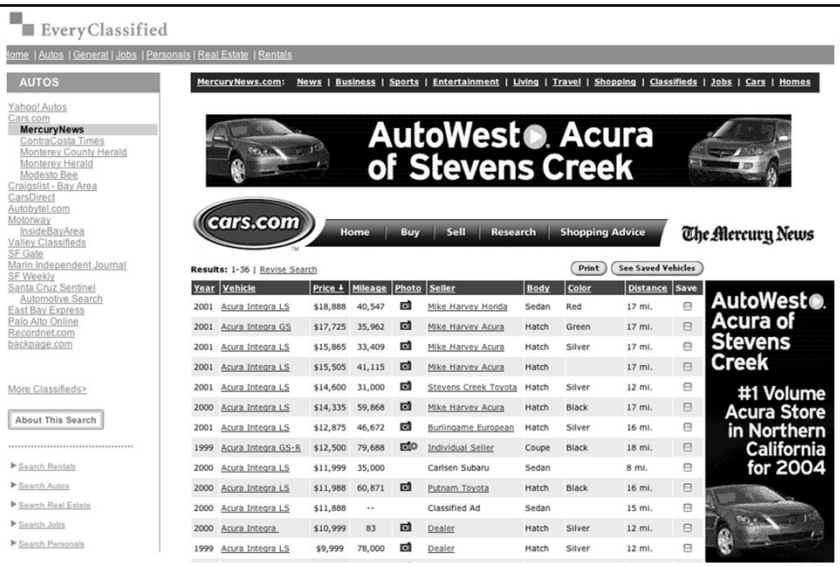


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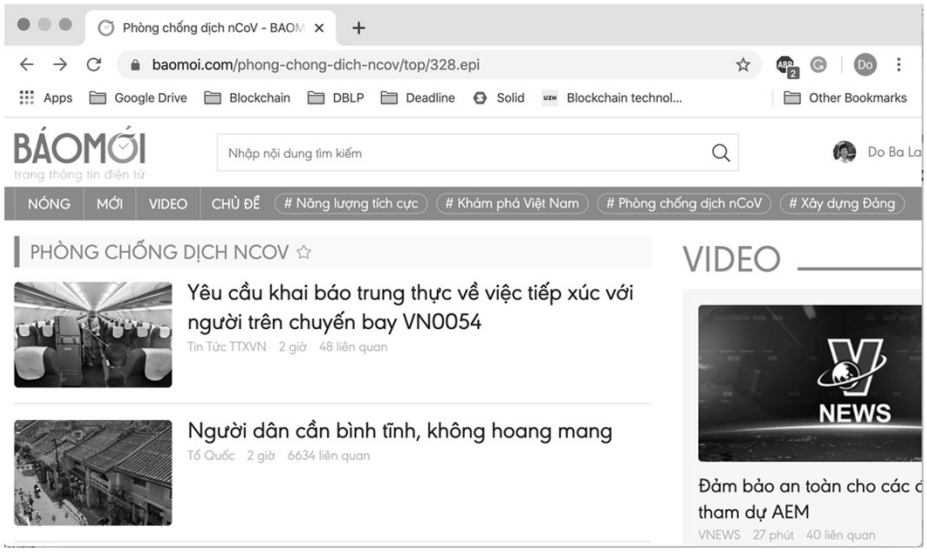
Create a single site to search for jobs/rentals/…

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Easily traverse between the site by clicking its name

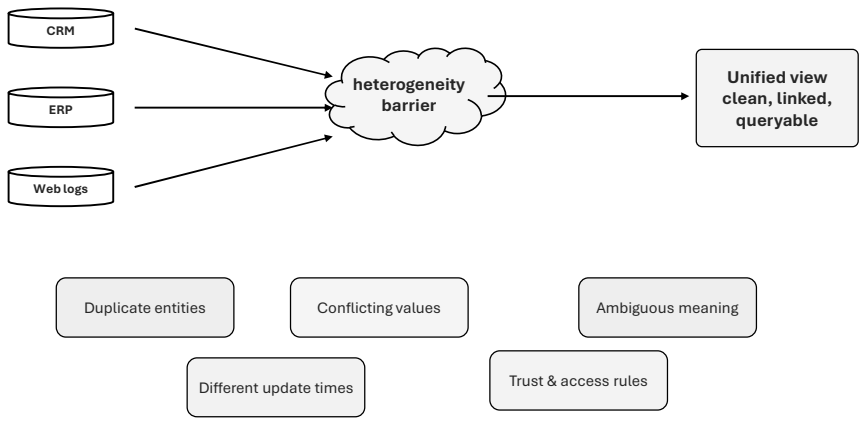
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Baomoi.com

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Heterogeneity



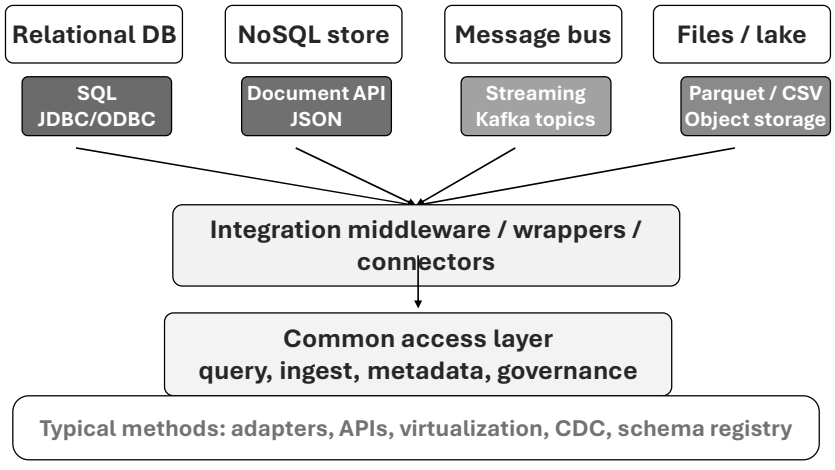
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Levels of heterogeneity

1	System / platform	Where and how data is stored or accessed
2	Syntactic / format	File formats, encodings, date and number syntax
3	Structural / schema	Tables, attributes, keys, nesting, granularity
4	Semantic / conceptual	Meaning of terms, units, business definitions
5	Instance / identity	Same real-world object appears differently
6	Quality / governance	Completeness, provenance, trust, permissions

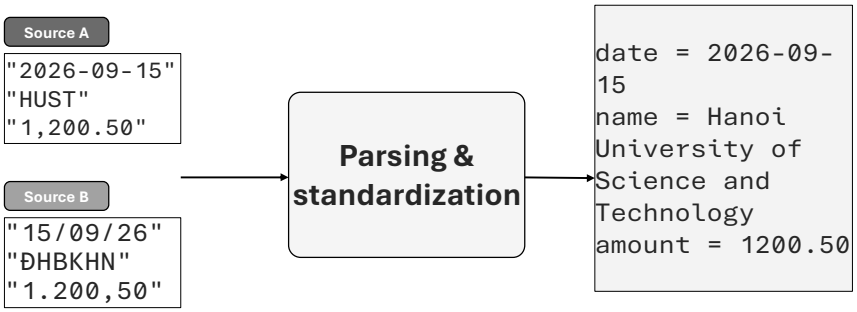
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System and platform heterogeneity



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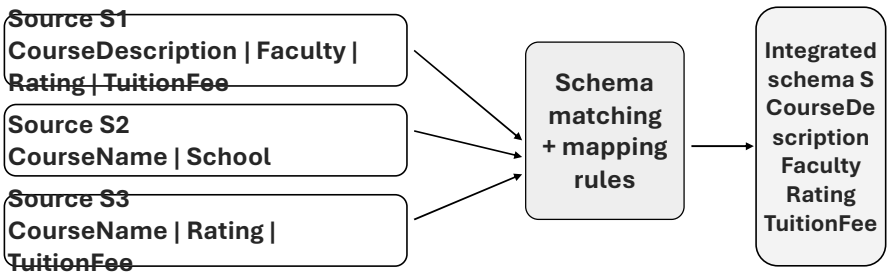
Syntactic and format heterogeneity



Common conflicts: file type, encoding, delimiters, date formats, decimal separators, abbreviations, languages.

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Structural and schema heterogeneity

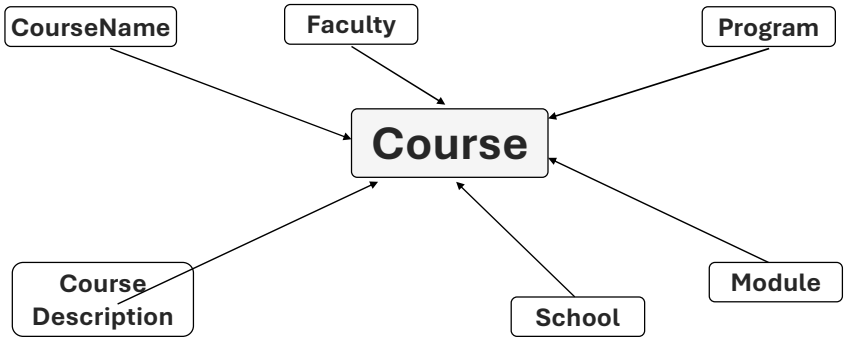


Structural conflicts: missing attributes, different normalization, nested vs flat, key mismatch, different granularity

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Semantic and conceptual heterogeneity

Resolution needs domain knowledge: ontology, controlled vocabulary, business glossary, expert validation.



Synonyms: Faculty ≈ School

Homonyms: “course” may mean a subject, training program, or degree path

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Data value and representation heterogeneity

Attribute	Source A	Source B	Integration action
TuitionFee	1200 USD	28,800,000 VND	currency conversion
Rating	4.7 / 5	92 / 100	scale normalization
Status	active	A	code dictionary
Missing	NULL	N/A / 999	missing-value policy

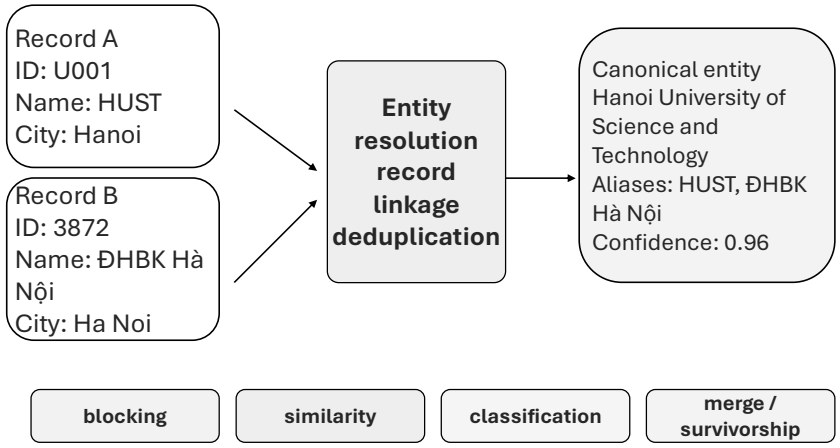
Goal: values should have one agreed type, unit, scale, and interpretation.

Typical techniques: data profiling, type inference, unit conversion, code mapping, outlier checks, standard reference data.

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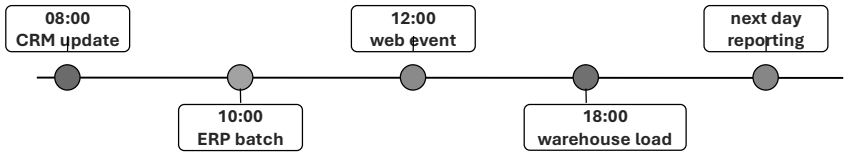
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Instance and identity heterogeneity



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Temporal and process heterogeneity



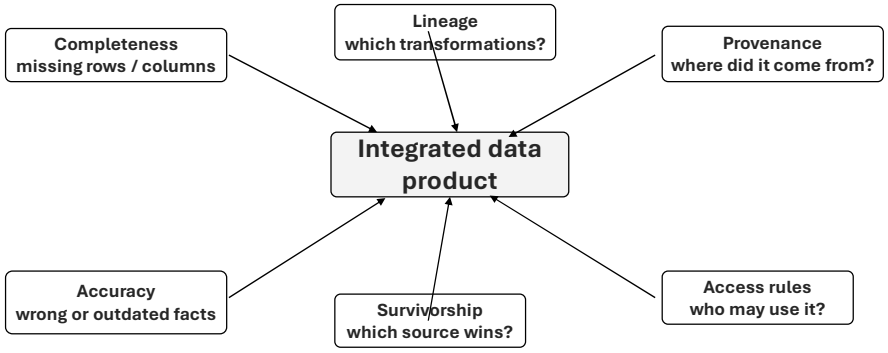
Freshness | **Snapshot consistency** | **Late-arriving data** | **Versioning**

Design choice: define whether queries use “latest available data” or a consistent time-based snapshot.

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Quality, provenance, and governance heterogeneity

Integration is not only about combining data; it must also preserve trust and usage constraints.



A technically correct integration can still be unusable if users cannot assess trust, lineage, and compliance.

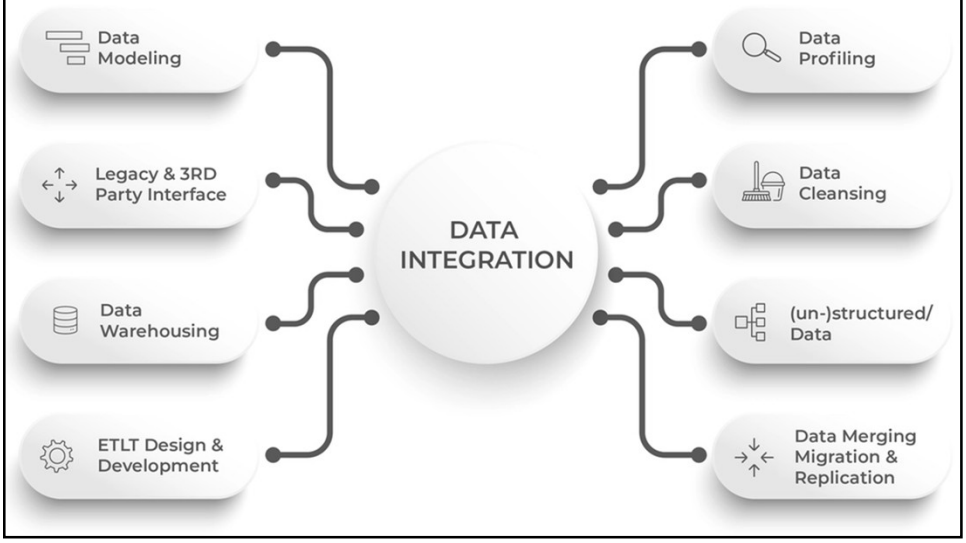
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Goal of Data Integration

- Providing an uniform query access to a set of data sources
- Handling with
  - Number of sources: from tens to millions
  - Different levels of heterogeneity

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Data Integration issues



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Outline

✓Data integration as a new abstraction
✓Examples of data integration applications
✓Heterogeneity
✓Goal of data integration
➢Data integration architectures
• Big Data Integration - Challenges

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Virtual, Warehousing and in Between

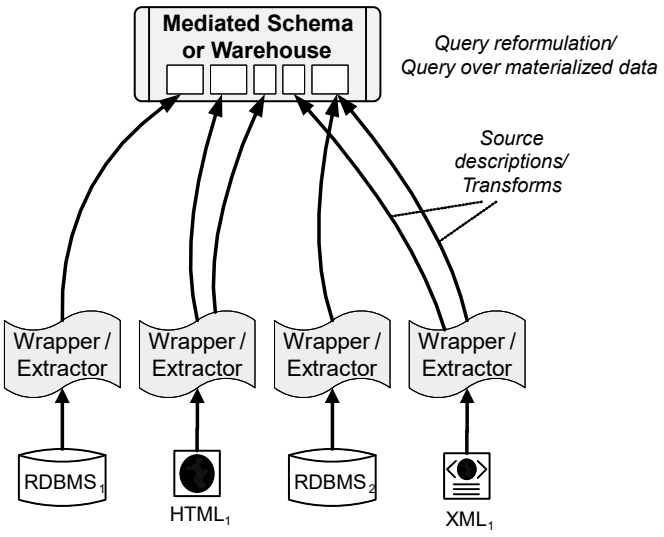
- Virtual data integration: leave the data at the sources and access it at query time.
- Data warehousing: integrate by bringing the data into a single physical warehouse

❖semantic heterogeneity

❖Numerous intermediate architectures.

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Virtual Data Integration Architecture



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Example



Movie(title, director, year, genre)
Actors(title, actor)
Plays(movie, location, startTime)
Reviews(title, rating, description)

S1 **Movies** (name, actors, director, genre)

S2 **Cinemas** (place, movie, start)

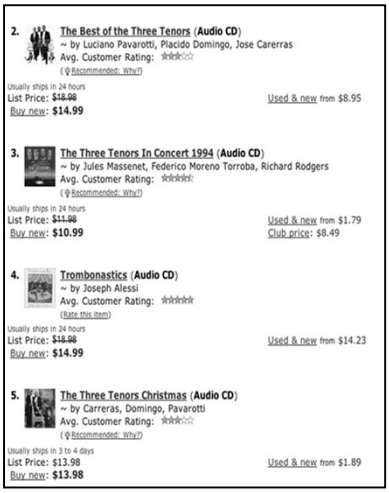
S3 **CinemasInNYC** (cinema, title, startTime)

S4 **CinemasInSF** (location, movie, startingTime)

S5 **Reviews** (title, date, grade, review)

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Wrappers



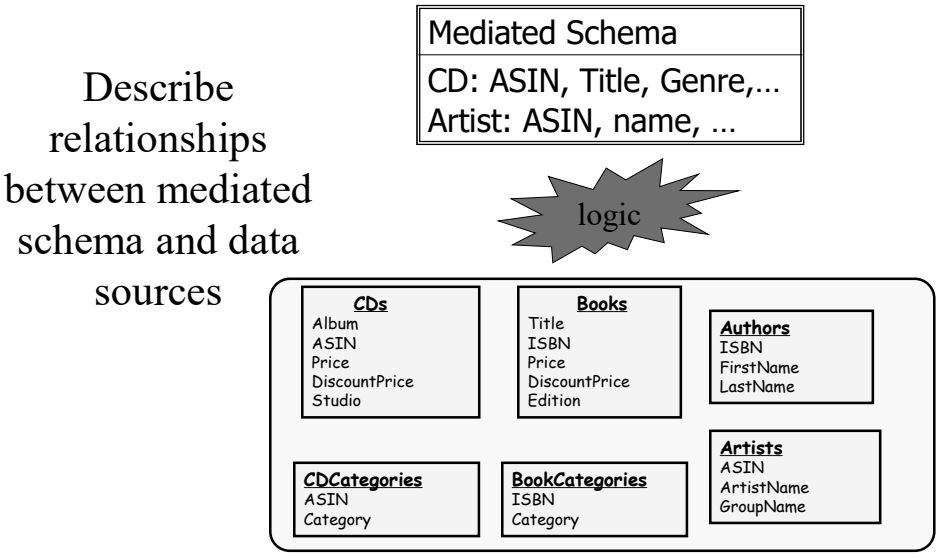
<cd> <title> The best of … </title>
<artist> Abiteboul </artist>
<artist> Pavarotti </artist>
<artist> Domingo </artist>
<price> 19.95 </price>
</cd>
…

Send queries to data sources and transform answers into tuples (or other internal data model).

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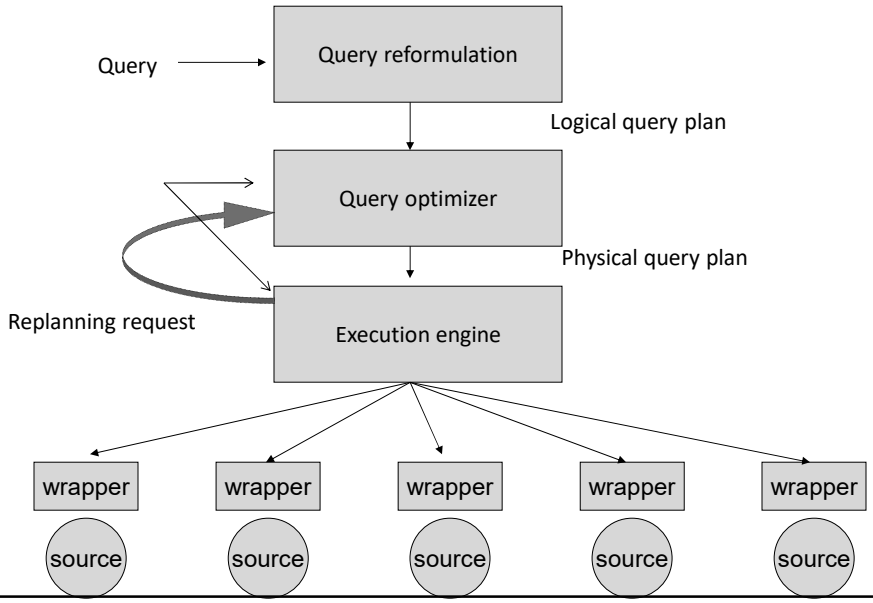
Mediation Languages

Describe relationships between mediated schema and data sources



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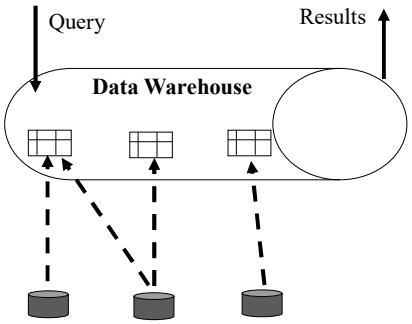
Query Processing



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Data Warehouses – Offline Replication

- Determine physical schema
- Define a database with this schema
- Define procedural *mappings* in an "ETL tool" to import the data and clean it.
- Periodically copy all of the data from the data sources
  - Note that the sources and the warehouse are basically independent at this point



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Pros and Cons of Data Warehouses

- ✗ Need to spend time to design the physical database layout, as well as logical
  - ✗ This actually takes a lot of effort!
- ✗ Data is generally not up-to-date (lazy or offline refresh)

- ✓ Queries over the warehouse don't disrupt the data sources
- ✓ Can run very heavy-duty computations, including data mining and cleaning

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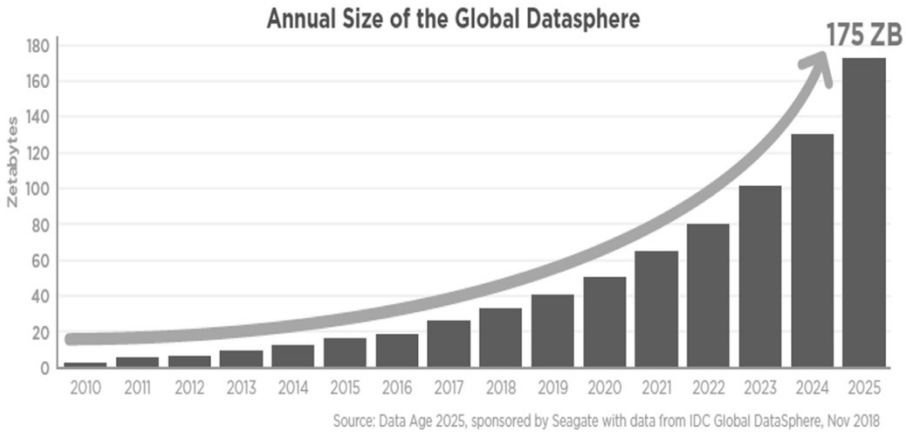
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- ✓Data integration as a new abstraction
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- ➢Big Data Integration - Challenges

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How big is big data?



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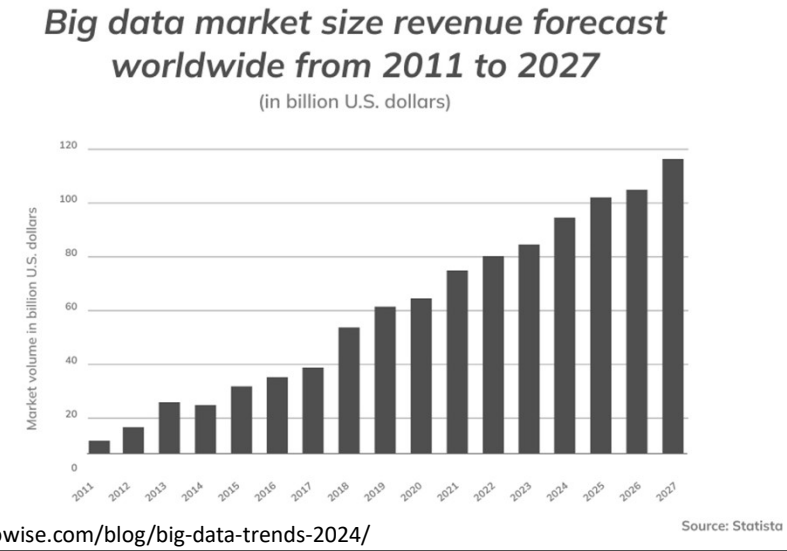
Big Data Growth Statistics

- An internet user generates ~ 1.7 megabytes (MB) of data / second.
- 2022: ~ 97 zettabytes - the estimated volume of data created worldwide
- 2023:
  - ~ 2/3 world population be online
  - internet users generate nearly 3 times the volume of data generated in 2019.
- 2025:
  - people will create more than 181 ZB of data. That's 181, followed by 21 zeros.
  - there will be 55.7 billion connected IoT devices. These IoT devices alone will generate almost 80 ZB by 2025.
- An internet user would need more than 180 million years to download all the data from the web.
- Nearly 80% of companies estimate that 50%-90% of their data is unstructured. Think text, video, audio, web server logs, or social media activities.

https://www.brimco.io/analytics/big-data-analytics-statistics/

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Big Data Market



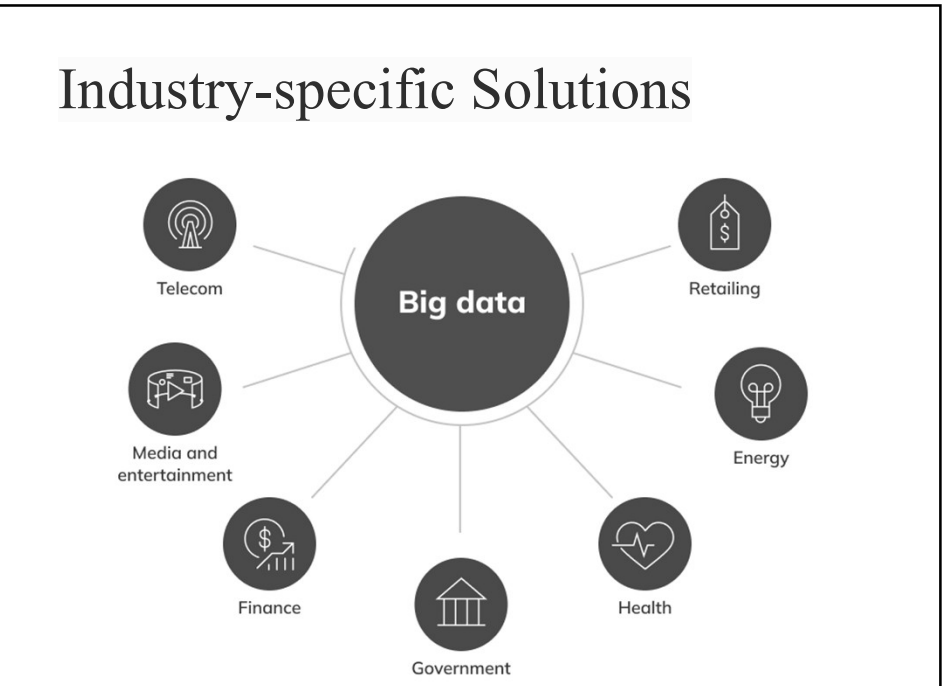
https://innowise.com/blog/big-data-trends-2024/

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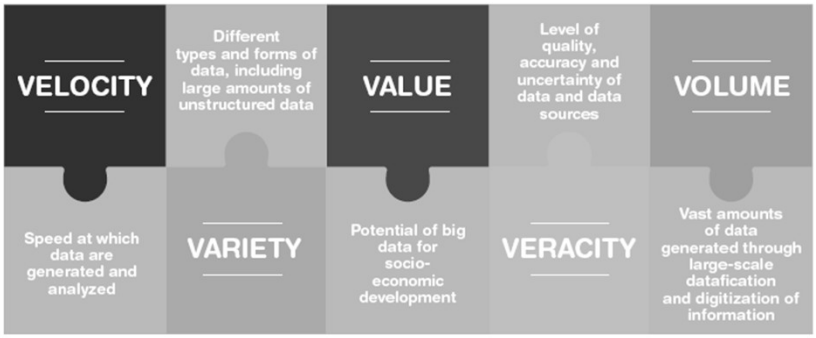
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https://innowise.com/blog/big-data-trends-2024/

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Big data 5'V

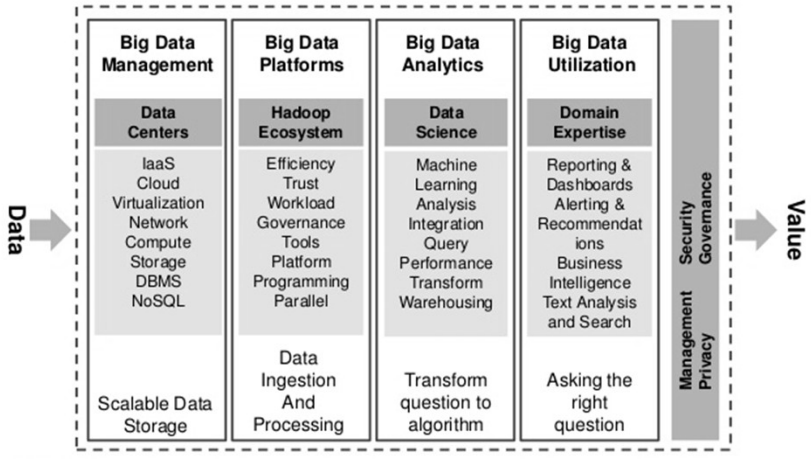


Big data is a term for data sets that are so large or complex that traditional data processing application software is inadequate to deal with them (wikipedia)

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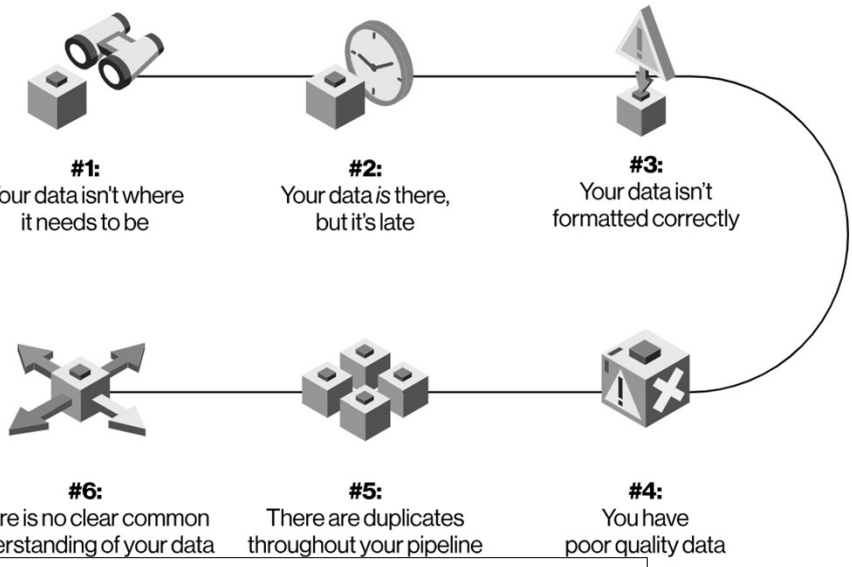
Big data technology stack



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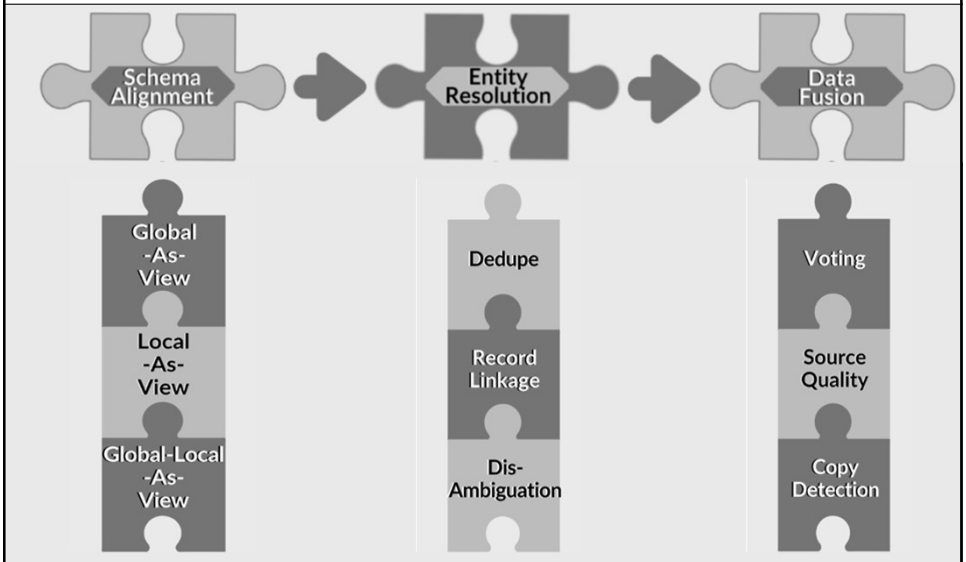
Challenges



https://www.cloverdx.com/blog/biggest-data-integration-challenges

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Task for Big Data Integration



https://www.selecthub.com/big-data-analytics/big-data-integration/

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